Shadows and residues in finite coset geometries

Human Compression project: private partial recovery

The purpose of this note is to decide geometric shadow intersection properties using the same finite coset actions in every residue. The orbit and witness reductions are those of Jacobs and Leemans [JL, Section 3], restated in the independently reviewed working note. The extension below makes the residue quantifiers and the input geometry tests explicit. It reuses the previously accepted flag-constrained covering proof and the panel-recognition arguments; it is not a new claim to the source algorithm.

1 Orbit and witness recognition

An incidence geometry has a finite set of types I, reflexive symmetric incidence, and no incident distinct objects of the same type. Every flag extends to a chamber. Suppose a finite group G acts transitively on flags of each type. Fix a chamber $C = \{p_i : i \in I\}$, use right actions, and write

$$H_i = G_{p_i}, \quad H_T = \bigcap_{t \in T} H_t, \quad H_\varnothing = G, \qquad C_T = \{p_t : t \in T\}.$$

The i-shadow $\mathrm{Sh}_i(F)$ is the set of type-i objects incident with every member of F. This geometric shadow property is distinct from the subgroup intersection property for distinguished generators of a C-group.

Lemma 1. *For every i, T, h we have*

$$\mathrm{Sh}_i(C_T^h) = (p_i^{H_T})^h.$$

Proof. If $i \in T$, both sides are the singleton $\{p_i^h\}$. Otherwise a type-i object y incident with C_T completes it to a flag of type $T \cup \{i\}$. Flag transitivity gives an element carrying $C_{T\cup\{i\}}$ to that flag while fixing each member of C_T. It belongs to H_T and carries p_i to y. Conversely every element of H_T fixes C_T, so its image of p_i is incident with C_T. Apply h to obtain the stated equality. □

The objects of type i are identified with right cosets $H_i\backslash G$. Thus the lemma uses one permutation action for each type. All shadows in that action are orbits of intersections of the chamber stabilizers. The recognition theorem in the current common theory for finite geometries can establish the required flag transitivity: for every $i \notin T$, the intersection $\bigcap_{t\in T} p_i^{H_t}$ must equal $p_i^{H_T}$. Induction on $|T|$ proves that this condition is sufficient, starting with the empty flag. It checks the entire set of eligible extensions, not a sample of chambers.

Theorem 2. *Fix a type i. The following condition is equivalent to the geometric shadow intersection condition at type i: for every element x and flag F, the intersection $\mathrm{Sh}_i(x) \cap \mathrm{Sh}_i(F)$ is empty or equals $\mathrm{Sh}_i(F')$ for a flag F' incident with both x and F.*

For each $j \neq i$, put $K = H_i \cap H_j$. For every nonempty $T \subseteq I \setminus \{i\}$, choose representatives h of

$$(H_i \cap H_T)\backslash H_i / K.$$

For each such representative form

$$X = p_i^{H_j} \cap (p_i^{H_T})^h.$$

If $|X| > 1$, there must be a subset $U \subseteq I \setminus \{i\}$ and an element $q \in K$ such that

$$(p_i^{H_U})^q = X, \qquad p_u^q \in (p_u^{H_T})^h \quad \text{for every } u \in U.$$

For a fixed U it suffices to take one representative of each right coset of $K \cap H_U$ in K.

Proof. Cases where x has type i, F contains type i, or F is empty have an immediate witness whenever the intersection is nonempty. Otherwise choose p in the nonempty intersection. The pair $\{p, x\}$ is a flag of types $\{i, j\}$; normalize it to $\{p_i, p_j\}$. The normalized F lies in the residue of p_i. Flag transitivity on $T \cup \{i\}$ expresses it as C_T^h with $h \in H_i$. Its stabilizer inside H_i is $H_i \cap H_T$. The remaining choices of normalization act by K, giving precisely the displayed double cosets. Lemma 1 identifies its intersection of shadows with X.

The singleton case has the witness consisting of its unique type-i object. If $|X| > 1$, a witness cannot contain type i. Let its type be U. Because $p_i \in X$, it is incident with p_i, and by assumption it is incident with p_j. Hence its union with $\{p_i, p_j\}$ is a flag. Flag transitivity expresses the witness as C_U^q for $q \in K$. Two elements of K give the same such flag exactly when they are in the same right coset of $K \cap H_U$. Lemma 1 gives the first displayed condition. Incidence with C_T^h says that, for every $u \in U$, its type-u object belongs to $\mathrm{Sh}_u(C_T^h)$, giving the second.

Conversely C_U^q is a flag. Since q fixes p_i and p_j, it is incident with those two objects. The second displayed condition makes it incident with C_T^h, and the first identifies its i-shadow with X. Transporting it back gives the required original witness. Every normalization and every orbit representative preserves these conditions, which proves sufficiency as well as necessity. □

There are two useful exact shortcuts. The candidate type U can be discarded unless $|p_i^{H_U}| = |X|$. Also, if p_j is incident with C_T^h and $|X|$ is the smaller of the two original shadow sizes, one of the original flags is already a witness. Both are consequences of the theorem and preserve its complete search space.

2 Every residue and every rank

For $J \subseteq I$, let R_J be the residue of the base flag C_J. Its types are $I \setminus J$. Empty residues are not tested for an existential type condition.

Theorem 3. *Suppose the coset system is flag transitive. The map*

$$(H_J \cap H_i)h \longmapsto H_i h, \qquad h \in H_J, \quad i \in I \setminus J,$$

is an isomorphism of the coset geometry inside H_J with R_J. In particular H_J is flag transitive on R_J. For each $k \in \{2, \ldots, |I|\}$ it suffices to test precisely the base residues R_J with $|I \setminus J| = k$. On each of them apply Theorem 2 with group H_J and parabolics $H_J \cap H_i$. All type tests in all these residues give $(\mathrm{IP})_k$*; at least one type test in each residue gives* $(\mathrm{WIP})_k$*. The conjunction over $k = 2, \ldots, |I|$ gives respectively IP and WIP.*

Proof. An object of type i incident with C_J extends that flag. Flag transitivity therefore writes it as $H_i h$ for $h \in H_J$. Two such representatives give the same object exactly when they give the same right coset of $H_J \cap H_i$ inside H_J. This proves object bijectivity. If the small cosets intersect, their common representative gives incidence of their images. Conversely, two incident images together with C_J form a flag. Flag transitivity gives a representative in H_J for both images, so their small cosets intersect. The same argument, applied to an arbitrary flag of the residue together with C_J, proves flag transitivity under H_J.

Every flag of type J is a translate of C_J, so all its residues are isomorphic with their type labels retained. This proves that one base residue per subset is sufficient; it does not identify different subsets or assume their residues are isomorphic. Inside a fixed residue the preceding

theorem is an equivalence at each remaining type. Universal quantification of those types is IP at that rank; the existential type for WIP is chosen separately in each residue. These are precisely [JL, Lemmas 2.1–2.2]. Rank-one shadow intersections are empty, singletons, or the whole sole type, with the corresponding immediate witnesses. Rank zero has no available type and is excluded, as in those lemmas. □

There are $[G : H_J]$ flags of type J, so the report records that many actual residues represented by each base residue. It computes every remaining type, even when one failed type would already refute IP. Thus its WIP output does not silently use a fixed global type.

3 Geometry hypotheses are separate checks

The supplementary rows are not accepted because their total matches a published table. They define coset incidence structures, which need not be flag transitive. We first check the orbit equality preceding Theorem 2 for all i, T. This condition is necessary by Lemma 1. For sufficiency, normalize an existing flag of type T inductively. Every object eligible to extend it belongs to the intersection of the singleton shadows, hence to the H_T orbit by the checked equality. It follows that each extended flag is a translate of the corresponding base flag. Induction covers every flag type, and in particular every flag extends to a chamber.

Write $K = H_J$ and $P_i = K \cap H_i$ for a residue. The following tests are the actual source hypotheses, with kernels of actions allowed.

Lemma 4. *For a flag-transitive coset geometry the following conditions are exact.*

1. *Firmness holds if and only if $[H_{I\setminus\{i\}} : H_I] \geq 2$ for every i.*
2. *Residual connectedness holds if and only if $\langle P_i : i \in I \setminus J\rangle = K$ for every residue of rank at least two.*
3. *Residual weak primitivity holds if and only if in each residue of positive rank at least one transitive action on $P_i\backslash K$ is primitive of degree at least two.*
4. *The local condition $(2T)_1$ holds if and only if, in every rank-one residue, its point stabilizer has exactly two orbits on its objects and there are at least two objects.*

Proof. The objects completing the base panel missing type i are the cosets of H_I in $H_{I\setminus\{i\}}$. Flag transitivity covers every panel, proving the first claim. For the second, a coset path starting at the base chamber can multiply its representative successively by elements of the P_i. Conversely each such multiplication is realized by a path through the object fixed by that subgroup. Hence the connected component of the base chamber is exactly the orbit of their join, and all components are its cosets. The argument is applied inside each K; it is valid for rank two as well as higher ranks. The third statement is the definition of weak primitivity applied to each residue. Equivalently P_i must be maximal proper in K: blocks containing the base coset correspond to intermediate subgroups. This equivalence does not require a faithful action. Finally a transitive action is two-transitive exactly when the stabilizer of a point is transitive on the other points. Its two orbits are then that point and its complement. Again this depends on the permutation image, not on a faithful ambient action. □

These are the panel and coset-path arguments already used in the accepted Hall–Janko census [D]. The implementation tests the entire action, not only subgroup orders or a presumed diagram.

4 Rank two without a diagram assumption

In a rank-two incidence structure the only nonautomatic shadow test compares the neighborhoods of two distinct objects on the same side. At a fixed shadow type i, it passes exactly when every such intersection has size zero, one, or the whole type-i set. Indeed a witness of type i has singleton shadow; a witness of the other type cannot be incident with both distinct objects of that type. The empty flag has the whole shadow set. These are all possible witness flags. In particular comparing an object with itself is automatic and is excluded from this pair test.

Both type tests hold if and only if the graph has no four-cycle or is complete bipartite. If a four-cycle is present, its two vertices on one side have at least two common neighbors, hence all vertices of the other side. Every pair on that other side consequently has at least two common neighbors and must meet the entire first side. This proves completeness. The converse follows immediately. This is the accepted direct proof [Cover], recovering the partial-linear-space/generalized-digon criterion used in [JL, Section 3]. We still record the two directional tests separately, and do not replace the weak condition by an assumed diagram.

For a coset action, fix the first object of the opposite type j. Representatives of $P_j \backslash K / P_j$ cover its possible second objects. Discard the identity double coset, which represents the same object. Compute the two neighborhoods by their actual coset-action orbits. Their intersections then perform the criterion just proved.

5 Bounded archive application

The parser reads only literal permutations (and the literal identity) from the supplementary archive. It removes comments before applying the last active reassignment of `geo`. It records both the construction index and the resulting active index. Neither is asserted to be a published geometry label. A commented selection with an empty placeholder in the rank-four Mathieu file is not executed.

For the Mathieu group M_{11}, the current complete checks cover thirteen active rank-three rows and eleven active rank-four rows. Every row passes all six source filters: flag transitivity, firmness, residual connectedness, RWPRI, $(2T)_1$, and $(\mathrm{IP})_2$. All residues of rank at least two are checked, not only the original geometries.

Input rank	tested residue rank	IP	weak only	not weak
3	2	13	0	0
3	3	6	1	6
4	2	11	0	0
4	3	2	0	9
4	4	2	0	9

The full rank-three IP rows in active archive order are $1, 3, 9, 10, 11, 13$; row 2 has WIP but not IP. In rank four, rows $1, 3$ have full IP and all other rows fail full WIP. The latter are construction rows $10, 2$. These aggregate values agree with the corresponding entries of Tables 1 and 3, but that agreement is a comparison after the computation, never an input filter or a proof of a published label. The source names the positive rank-four geometries $4.1, 4.4$ in its earlier classification; that exact identification remains pending.

A failed type test records the actual normalizing permutation, flag types and intersection. Every possible witness type and its stabilizer cosets are then exhausted by the checker. Positive decisions exhaust every required double-coset orbit. The GAP finite-group routines are an explicit trusted computational input; no Lean formalization of these catalogue checks is claimed.

6 The active M12 files and their filters

The other bounded batch covers all twenty active rank-three rows, all twenty active rank-four rows, and the single rank-five row of the M_{12} archive. These are the literal final active lists, not the numbers assumed from the printed tables. Their groups are explicitly conjugate to the named natural Mathieu group; the computed conjugating permutations are retained with the input binding.

Every row passes flag transitivity, firmness, residual connectedness, RWPRI and $(\mathrm{IP})_2$. Local two-transitivity rejects the rank-three active rows $15, 17, 20$ (construction rows $27, 29, 33$), and rank-four active rows $4, 7$ (construction rows $25, 23$). For rank-three row 17, the relevant panel action has degree 11 and order 55: its point stabilizer has orbit lengths $1, 5, 5$. The other failures have degree-three cyclic panel images, whose point stabilizers have three singleton orbits. These explicit failures do not depend on interpreting an archived row as a named published geometry. The two-transitivity test is for the specified G action; no claim about a larger full automorphism group repairs it.

After these actual filters, the computed counts are as follows.

Input rank	residue rank	eligible rows	IP	weak only	not weak
3	2	17	17	0	0
3	3	17	7	1	9
4	2	18	18	0	0
4	3	18	4	0	14
4	4	18	3	0	15
5	2	1	1	0	0
5	3	1	1	0	0
5	4	1	1	0	0
5	5	1	0	0	1

In particular the rank-five example satisfies every proper-residue condition, but fails WIP in its full rank. This is why a computation of proper residues alone cannot certify the full property.

The source's rank-three summary has eighteen rows, with counts $7, 1, 10$; the specified active archive under its stated action filters has seventeen, with counts $7, 1, 9$. This discrepancy is retained. The earlier phase-six archive comparison independently flags exactly these same three rank-three and two rank-four inputs as non-$2T$. Its independent broader census also records an omitted non-$2T$ rank-three class; adding that class would not repair this filtered seventeen-versus-eighteen difference. We reuse that prior evidence without repeating its subgroup census or claiming a newly discovered literature correction. The rank-four and rank-five aggregate shadow columns agree with the printed source, but exact published row labels still require separate identification.

7 Attribution and remaining boundary

The supplementary lists and orbit algorithm originate in [JL]. The present application does not reprove completeness of the source's sporadic geometry classifications. The entire source contract, the remaining entries of Tables 1–8, their exact published labels, Section 5 on Suzuki and inductively minimal geometries, and its complete software correspondence remain pending. In particular active archive sizes need not equal the published filtered numbers: the two Mathieu M_{12} files each select twenty rows (with the exclusions just proved), and the supplied J_2 rank-four file has one row although the table has three. Our earlier accepted full Hall–Janko census provides the three rank-four witnesses independently; its large census need not be rerun. The executable Magma listing in source Section 6 also requires a matched software boundary before

any numerical comparison; it is not silently counted only on the source side. No whole-original, question or numerical-compression credit is claimed.

References

[JL] P. Jacobs and D. Leemans, *An algorithmic analysis of the intersection property*, LMS J. Comput. Math. 7 (2004), 284–299, fixed primary PDF and its supplied supplementary archive.

[Cover] Human Compression project, *Geometric intersection as a flag-constrained covering problem*, accepted `phase4_shadow_ip.md` and distinct audit. The finite covering proof is general; its original audit does not certify this new census.

[D] Human Compression project, *Stabilizers, extensions and the enumeration of finite geometries*, accepted Topic D common theory, panel recognition and Hall–Janko census. Source group catalogues and the accepted finite census remain explicit retained inputs.